



Vijayaragavan S 251801381-2025-AIDS ▾

V2

Started on	Tuesday, 15 September 2026, 9:31 AM
State	Finished
Completed on	Tuesday, 15 September 2026, 9:34 AM
Time taken	3 mins 36 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  int hasTwoSum(int arr[], int left, int right, int x) {
3      if (left >= right)
4          return 0;
5      int sum = arr[left] + arr[right];
6      if (sum == x) {
7          printf("%d\n%d\n", arr[left], arr[right]);
8          return 1;
9      }
10     if (sum < x)
11         return hasTwoSum(arr, left + 1, right, x);
12
13     return hasTwoSum(arr, left, right - 1, x);
14 }
15 int main() {
16     int n, x;
17     if (scanf("%d", &n) != 1) return 0;
18     int arr[n];
19     for (int i = 0; i < n; i++) {
20         scanf("%d", &arr[i]);
21     }
22     scanf("%d", &x);
23     if (!hasTwoSum(arr, 0, n - 1, x)) {
24         printf("No\n");
25     }
26     return 0;
27 }
```

	Input	Expected	Got	
✓	4 2 4 8 10 14	4 10	4 10	✓
✓	5 2 4 6 8 10 100	No	No	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)